

Exploratory Data Analysis Report

Task Management Dataset

This report presents the exploratory analysis completed in the supplied Jupyter Notebook. It consolidates the data overview, cleaning step, five required visualizations, and business observations.

1. Introduction

The dataset contains task-level information for a project/task management environment. Each record represents a task and includes task ID, task name, project ID, assigned employee, deadline, progress percentage, and task status. The purpose of the EDA is to understand task status, progress, project workload, employee allocation, and deadline risk.

2. Dataset Overview

Metric	Result
Total Tasks	500
Columns	7
Missing Values	0
Duplicate Records	0
Unique Projects	50
Unique Employees	50

3. Data Cleaning and Preparation

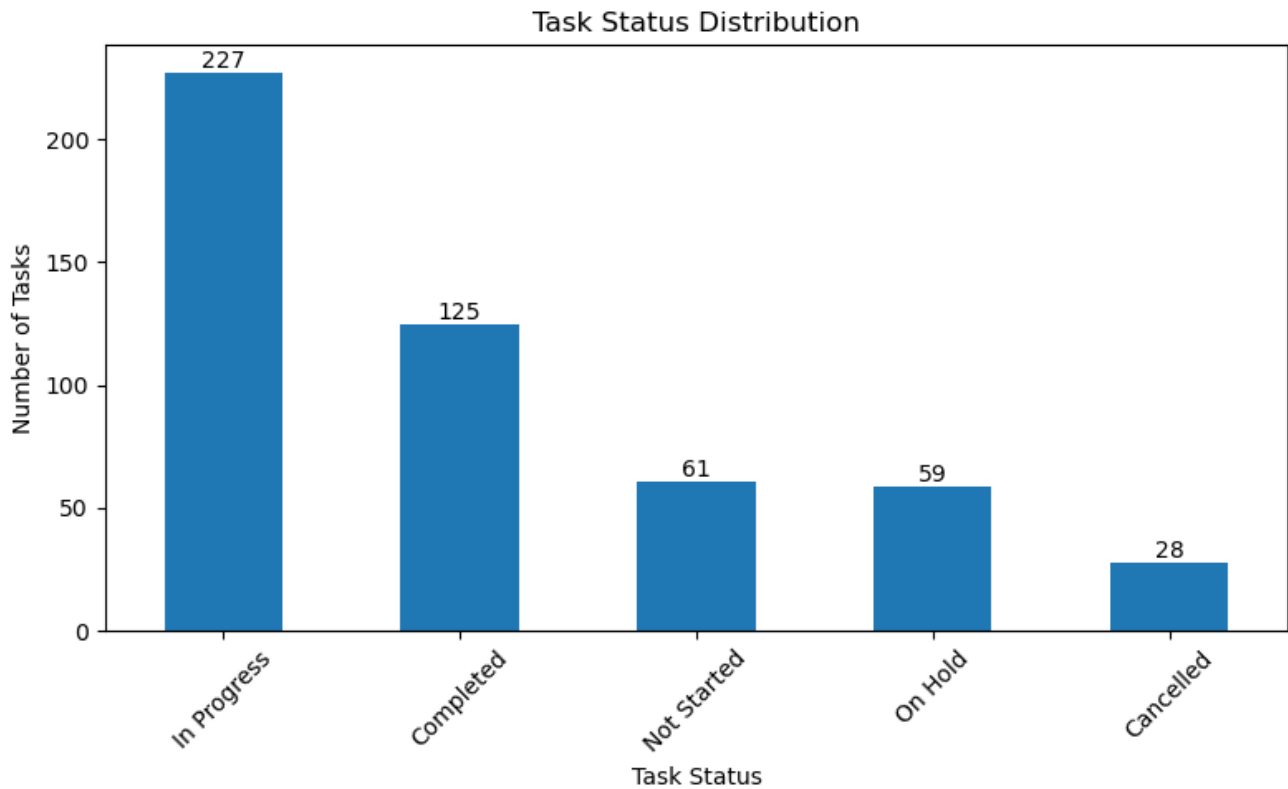
The notebook confirms that the dataset contains 500 non-null records across all seven columns. No missing values and no duplicate records were found. The **deadline** column was converted from object/string format to **datetime64** format so that deadline-based calculations could be performed.

4. Data Overview Findings

The dataset contains 500 tasks distributed across 50 projects and 50 employees. Because the missing-value count and duplicate-record count are both zero, no record-removal step was required for these two data-quality issues.

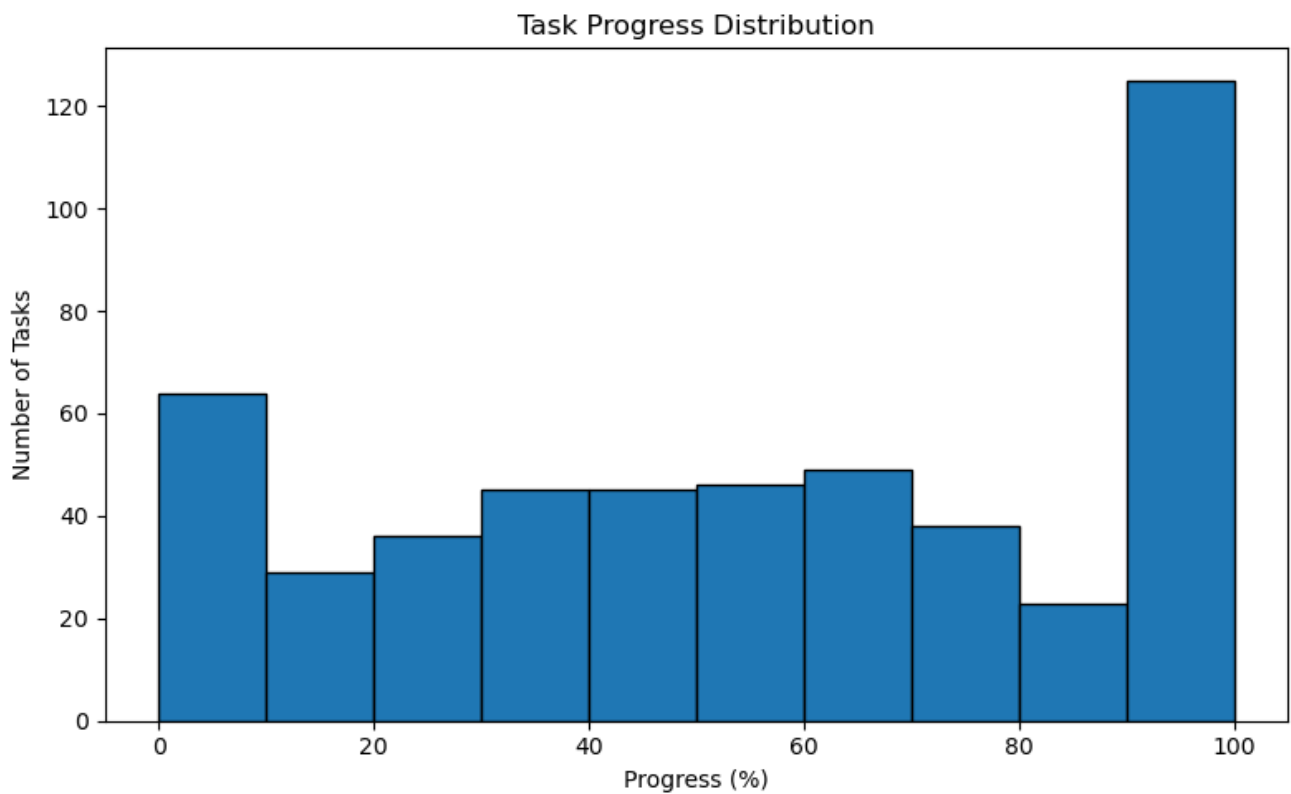
5. Required Visualizations

Task Status Distribution



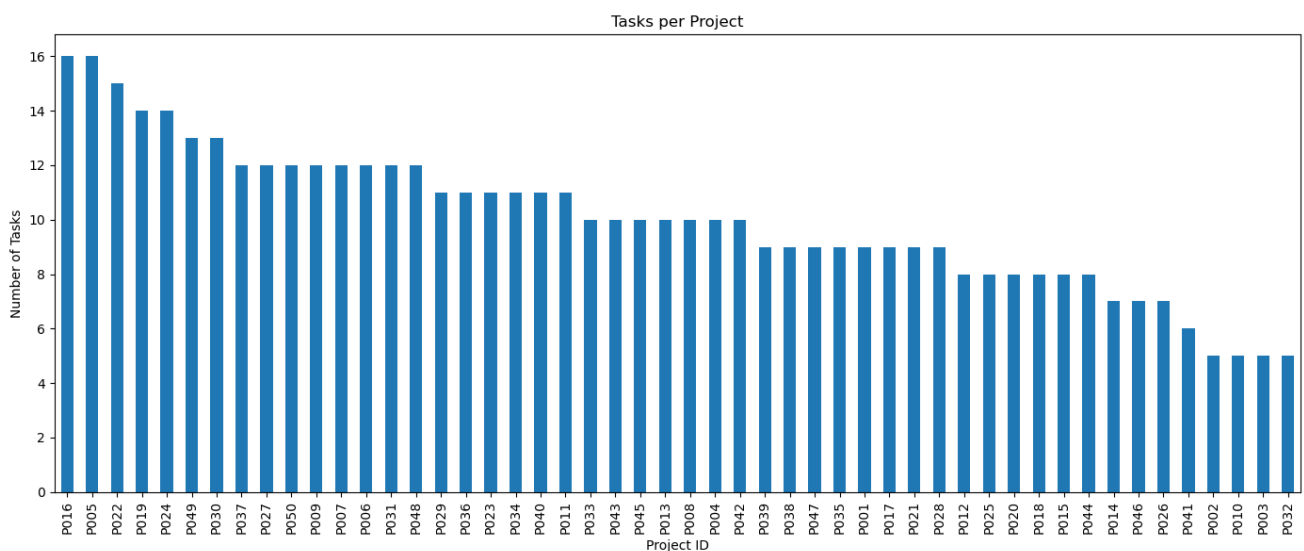
In Progress is the largest status category with 227 tasks, followed by Completed with 125. The remaining tasks are distributed across Not Started, On Hold, and Cancelled statuses.

Task Progress Distribution



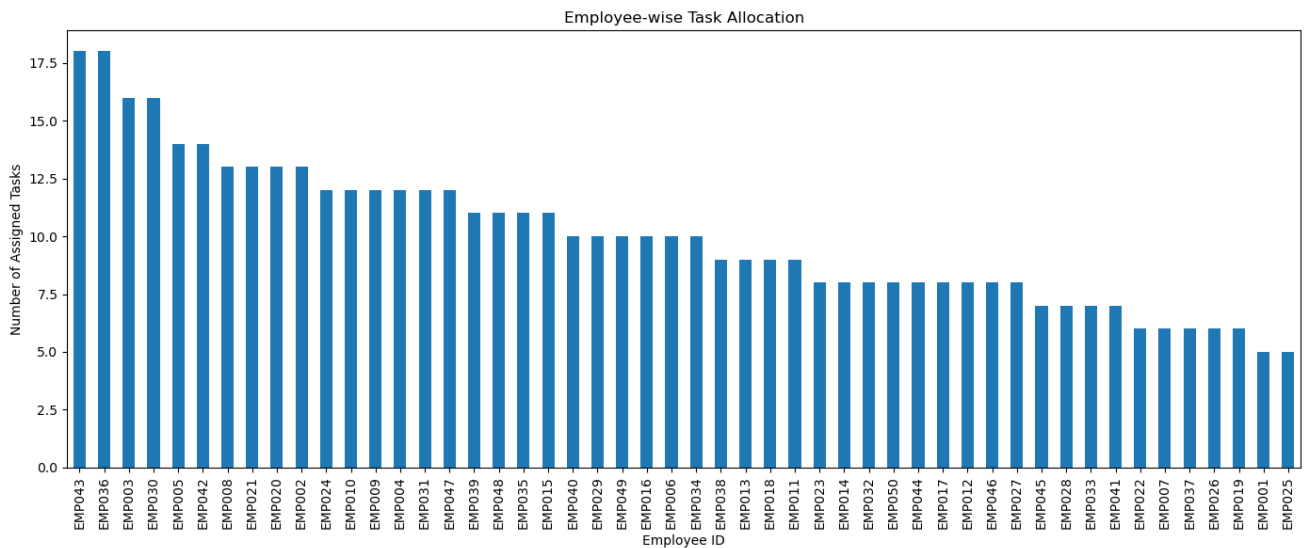
The histogram shows how task progress percentages are distributed from 0% to 100%, providing a view of how far tasks have progressed.

Tasks per Project



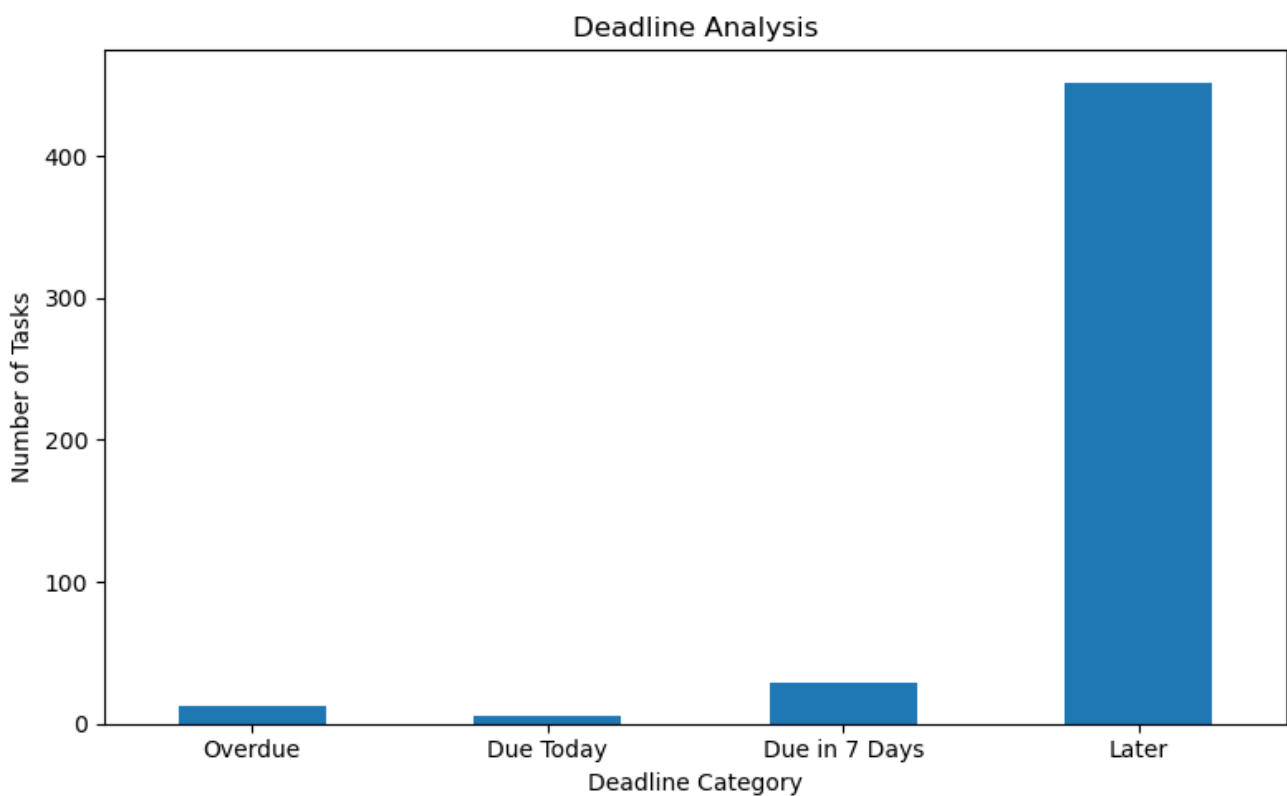
The project-level chart compares the number of tasks assigned to each project and helps identify projects with relatively high task volumes.

Employee-wise Task Allocation



The employee-level chart shows how tasks are distributed among employees and helps identify differences in assigned workload.

Deadline Analysis



As of 29 August 2026, most tasks fall into the Later category, while a smaller group is due within seven days, due today, or already overdue.

6. Business Observations

Observation 1 — High number of tasks are in progress

227 out of 500 tasks are currently In Progress. This makes In Progress the largest task-status category and indicates that a substantial amount of work is still ongoing.

Observation 2 — Completion rate is 25%

125 of the 500 tasks are Completed, giving a completion rate of 25%. A larger number of tasks remain In Progress, Not Started, or On Hold.

Observation 3 — Employee workload is not completely balanced

EMP043 and EMP036 each have 18 assigned tasks, while other employees have fewer tasks. This indicates differences in task allocation and suggests that workload distribution can be monitored.

Observation 4 — Some projects have higher task volumes

Projects P016 and P005 have the highest task volume at 16 tasks each. These projects may require closer monitoring and resource planning.

Observation 5 — Deadline risk exists

As of 29 August 2026, the notebook identifies 12 unfinished tasks that are already overdue. These tasks represent a potential schedule-risk area and may need immediate attention.

7. Conclusion

The EDA provides a clear snapshot of the task management dataset. The main areas for management attention are the high number of ongoing tasks, the relatively low completion rate, differences in employee workload, higher task volumes in specific projects, and overdue unfinished tasks. These findings can support better workload balancing, project monitoring, and deadline management.